BASKETBALL MANAGER

Project Overview & Technical Reference

Author	Arya Chakraborty	**Engine**	Godot 4.6.2 (GDScript)
Platform	PC (macOS / Windows / Linux)	**Database**	SQLite (godot-sqlite)
Started	April 2026	**Status**	**In Development — Match Engine & Season Loop Complete**

1. Vision & Concept

Basketball Manager is a deep, simulation-based basketball management game inspired by Football Manager and Out of the Park Baseball. The player takes the role of a head coach, managing every aspect of a basketball program across multiple interconnected leagues.

Unlike existing basketball management games, this aims for Football Manager depth: a rich simulation engine that evaluates 53 player attributes per action, a living world where AI teams make intelligent decisions, and a career mode where the coach's reputation grows organically from G League gyms to NBA arenas.

Core Design Pillars

Simulation depth	Every match outcome traces back to 53 individual player attributes across 7 categories
A living world	AI teams trade, sign free agents, draft players, and develop rosters independently
Multi-league ecosystem	NBA, G League, college, and overseas leagues — all connected and config-driven
Career progression	Coach reputation gates job opportunities; build from obscurity to dynasty
No 3D match engine	Text-based simulation with detailed play-by-play commentary
Coaching influence	Philosophy, defensive schemes, mismatch hunting, and rotation management all shape the sim
Inbox/news system	FM/2K-style message inbox — every significant event lands as a readable message

2. Leagues & Scope

The player selects which leagues to include when starting a new game. Each league is config-driven via JSON and can be toggled independently.

NBA (30 teams)	Fully playable. Full salary cap system, luxury tax, max contracts, two-way deals, 15-man roster, G League affiliates.
G League (30 teams)	Full team entities with own identity, standings, and coaching career path. 1-year flat contracts. Inherits NBA affiliate owner personality.
College Basketball	Recruiting pipeline into the draft. Fictional programs. 4-year player development arc.
EuroLeague / Overseas	International scouting layer. Players can sign abroad, be scouted, and imported via draft or free agency.

3. Game Modes

There is one game mode: Career Mode. The player creates a coach and picks any team for their first job regardless of reputation. All subsequent jobs are governed by reputation, openings, and interviews — exactly like FM. Staying at a team requires performing well enough to keep the job. There is no single-team lock-in mode.

Manager Creation

Manager creation shapes the coach's starting reputation, coaching attributes, and tendencies. It consists of five steps.

Step 1 — Playing Career

Option	Description	Rep Bonus
NBA Legend	Defined an era, won championships, highest individual honours	+40
NBA Veteran	Played significant minutes, respected career	+30
NBA Journeyman	Reached the league, bounced between rosters	+20
G League Pro	Career spent primarily in the G League	+12
College Star	Standout college player, never made the pros	+8
Overseas Pro	Played professionally abroad	+8
No Playing Career	Never played professionally	+0

Step 2 — Non-Playing Background

A recognition multiplier (Global ×1.5, National ×1.2, Regional ×1.0, Local ×0.7) is applied to the bonus.

Role	Sub-roles	Rep Bonus

Coaching	Head Coach, Assistant Coach, Player Development Coach, Skills Trainer	+5 to +15
Front Office	Scout, GM Assistant, Analytics Director	+5 to +10
Media & Broadcasting	Analyst, Commentator, Pundit	+3 to +8
Medical & Performance	Athletic Trainer, Strength & Conditioning Coach	+2 to +5
None	No non-playing background	+0

Step 3 — Coaching Qualifications

Qualification	Rep Bonus	Coaching Attr Impact
USA Basketball Gold License	+20	All coaching attrs Good–Very Good
USA Basketball Silver License	+12	Most attrs Average–Good
USA Basketball Bronze License	+6	Attrs Average
No Qualifications	+0	Attrs Poor–Average

Step 4 — Coaching Style (pick up to 3)

Fewer choices = stronger focus (full attribute bonuses). Three choices = versatile but diluted (bonuses at 60%).

Style	Philosophy	Attribute Effect
Offensive Mastermind	Build systems that maximise scoring	+Offensive Systems, +Tactical Flexibility
Defensive Architect	Stopping the other team is everything	+Defensive Systems, +Tactical Flexibility
Player Developer	Unlocking potential is your gift	+Player Development, +Working With Youth
Pace & Space	Fast, spread the floor, three-point heavy	+Offensive Systems, pace_target +0.15
System Builder	Rigid scheme, everyone fits the system	+Defensive Systems, +Tactical Flexibility
Players' Coach	Relationships drive everything	+People Management, +Motivating
Analyst	Data-driven decisions at every level	+Tactical Flexibility, +Scouting; hunt mismatches always on
Entertainer	Fans pay to watch your teams play	+Offensive Systems, pace_target +0.10

Step 5 — Personality (pick up to 2)

Personality	Description	Attribute Effect
Driven	Relentless focus on winning	+Authority, +Determination

Disciplined	High standards, accountability above all	+Authority; switch_on_screens = false
Players' Advocate	You fight for your players	+People Management, +Motivating
Resilient	Bounces back from adversity	+Determination, +Adaptability
Inspirational	Elevates everyone around you	+Motivating, +Authority
Accomplished	Thrives at any level in any situation	+Adaptability; reputation ×1.1

Coaching Attributes

Stored as integers 1–5, displayed descriptively (Poor / Average / Good / Very Good / Outstanding). Derived sim-facing float values are computed at match time from these ratings.

Offensive Systems	Quality of offensive schemes; determines philosophy weight accuracy in PossessionSim
Defensive Systems	Quality of defensive schemes; governs assignment correctness and help rotation breakdown rate
Player Development	Accelerates player growth; multiplier on PlayerDev tick
Scouting	Scout report accuracy; affects noise range on scouted attributes
Motivation	Pre/post-game team talk effectiveness; flat attribute bonus at match start
Tactical Flexibility	Mismatch detection rate and exploitation weight; mid-game adjustment quality
Fitness & Conditioning	Player stamina drain modifier; reduces fatigue accumulation rate
Working With Youth	Development bonus for players under 23
Adaptability (mental)	Adjusting to new rosters and situations
Authority (mental)	Locker room control; player discipline threshold
Determination (mental)	Keeps pressure from affecting coach performance
People Management	Player relationships; morale impact scale
Motivating (mental)	In-game player shouts effectiveness

Starting Reputation Formula

```
reputation = playing_career_bonus
           + (non_playing_bonus × recognition_multiplier)
           + qualification_bonus
           × personality_multiplier  -- 1.1 if Accomplished, else 1.0
```

0–20	G League assistant or head coach at lower level
21–35	G League head coach

36–50	NBA assistant / first-time NBA head coach candidate
51–65	Mid-tier NBA head coach
66–80	Top-tier NBA job candidate
81+	Contender-level NBA teams available from day one

Note: First job can be any team regardless of reputation. All subsequent jobs are governed by reputation and openings.

4. Technical Architecture

The project follows a strict separation of concerns: simulation logic never touches UI code, and all game state flows through global singletons (autoloads).

Layer Overview

UI Layer (Godot scenes)	Reads from WorldState and emits signals via EventBus. No simulation logic.
Autoloads (global singletons)	EventBus, WorldState, Database, Clock, GameManager — always in memory.
Simulation Engine (GDScript)	Pure logic classes: MatchEngine, PossessionSim, ShotResolver, RotationManager, SeasonScheduler, QuickSim.
Entity Layer (GDScript classes)	Data objects: Player, Team, Coach, Contract. Loaded into WorldState at game start.
Systems Layer	LeagueLoader, ScoutingSystem, NewsEngine — orchestration and cross-cutting concerns.
Data Layer (SQLite + JSON)	SQLite for saves and runtime data. JSON configs define league rules and structure.

Autoload Singletons

EventBus.gd	Signal hub. All cross-system communication goes through here. No direct node references.
Database.gd	SQLite connection, schema creation, all query helpers. All DB access goes through this.
WorldState.gd	Master game state: teams, players, coach, standings, schedule, cap figures, TV deal tracking.
Clock.gd	Advances game calendar. Triggers season milestones. is_match_day() reads schedule directly.
GameManager.gd	New game setup (calls LeagueLoader + ScheduleGenerator), save/load, screen transitions.

File Structure

```
basketball-manager/
```

```
  addons/godot-sqlite/              ← SQLite plugin
  autoloads/                        ← EventBus, Database, WorldState, Clock, GameManager
  core/entities/                    ← Player.gd, Team.gd, Coach.gd
  core/engine/match/                ← MatchEngine, PossessionSim, ShotResolver,
                                       RotationManager, CoachTactics, QuickSim,
CommentaryEngine
  core/engine/season/               ← ScheduleGenerator, SeasonScheduler, PlayoffEngine
  core/engine/personnel/            ← DraftEngine, TradeEngine, FreeAgency,
ContractGenerator
  core/engine/development/          ← PlayerGenerator, PlayerDev, InjuryEngine
  core/systems/                     ← LeagueLoader, ScoutingSystem, NewsEngine
  data/leagues/nba/                 ← 5 JSON config files
  data/leagues/g_league/            ← g_league_teams.json
  data/names/                       ← Procedural name banks
  ui/screens/                       ← Hub, Squad, MatchDay, Draft, Scouting
  ui/components/                    ← Reusable UI widgets
  assets/fonts/
```

5. Player Attribute Model

Players have 53 attributes across 7 categories, rated 1–20 internally. Scouts see inaccurate versions based on scout_knowledge (0.0–1.0). Hidden attributes are never exposed directly — they surface as observable patterns in player behaviour and performance.

Attribute Categories

Scoring (13)	Close Shot, Layup, Driving Dunk, Standing Dunk, Post Hook, Post Fade, Post Control, Mid-Range, Three-Point, Free Throw, Draw Foul, Hands, Shot IQ
Playmaking (6)	Ball Handling, Speed With Ball, Pass Accuracy, Pass Vision, Pass IQ, Off-Ball Movement
Defense (8)	Perimeter Defense, Interior Defense, Help Defense IQ, Pass Perception, Steal, Block, Defensive IQ, Defensive Consistency
Rebounding (2)	Offensive Rebounding, Defensive Rebounding
Athleticism (7)	Speed, Agility, Strength, Vertical, Stamina, Hustle, Overall Durability
Mental (9)	Anticipation, Composure, Concentration, Decision Making, Determination, Flair, Leadership, Teamwork, Work Ethic
Hidden (9)	Coachability, Consistency, Natural Fitness, Versatility, Adaptability, Injury Proneness, Important Matches, Concentration, Potential

Key Design Decisions

Scout visibility	44 visible attrs returned with noise ±6 at 0.0 knowledge, ±0 at 1.0. 9 hidden attrs never returned.
Delta-based potential	Each attribute has its own hidden growth ceiling. potential_deltas[attr] = how many more points it can grow.

Injury Proneness	High = bad (FM model). Not shown to scouts. Surfaces only through injury frequency patterns.
Important Matches	Hidden. High = rises to occasion. Low = nerves under pressure. Drives clutch modifier and superstar trade demand.
Flair	Visible. Drives improvised plays (no-look passes, euro steps, alley-oops) in PossessionSim. Composure gates risk of failure.
OVR rating	Position-weighted display only. Simulation always uses individual attributes directly.

6. Player Generation System

PlayerGenerator.gd produces realistic players via a multi-layer pipeline. The same generator serves NBA rosters, G League rosters, and draft classes.

Generation Pipeline

1. Roll tier	Weighted by league context. NBA: 20% Fringe / 40% Rotation / 28% Starter / 10% Star / 2% Superstar.
2. Roll archetype	Probability by tier: 20% Fringe → 100% Superstar. Archetype = primary skill + secondary skill.
3. Roll primary skill	Weighted by position. PG: 30% Passing, 25% Shot Creating, 20% Driving, 15% Three-Point, 10% Defending.
4. Roll secondary	Same pool as primary; primary skill gets 0.6× weight so same/same combos are allowed but not dominant.
5. Generate attrs	tier_mean + position_bias + primary_skill_bias + (secondary_bias × 0.6) + normal_noise(σ=2.0). Clamped 1–20.
6. Hidden attrs	Flat distribution: mean 8, σ=2.5. Fully independent of tier and position.
7. Age degradation	Physical attrs -0.6/year past age 26, capped -6. Mental attrs +0.3/year past 30, capped +3.
8. Potential deltas	Per-attribute growth ceilings. Younger + lower tier = more room to grow. Superstars near ceiling already.

7. Salary Cap & Contract System

Full NBA salary cap fidelity. All cap figures stored as percentages of the current cap so they automatically scale each offseason when the cap increases.

Cap Structure

Starting cap	$136,000,000 (2025 season)
Luxury tax threshold	Cap × 1.10

First apron	Tax threshold × 1.053 — restrictions on MLE and trades
Second apron	Tax threshold × 1.105 — hard restrictions, pick limitations
Annual cap growth	6% base rate ± 2% variance each offseason
TV deal spike	Every 8–10 seasons: 20–30% cap jump. Telegraphed 2 seasons in advance via inbox.

Contract Types

Max contract	25%/30%/35% of cap based on experience. Up to 5 years.
Rookie scale (R1)	4 years, years 3–4 team options. Slot-based % of cap. 8% annual escalation.
Second round	2 years, year 2 team option. Veteran minimum range.
Two-way	$600K flat. Does not count against cap. 50 NBA day limit.
G League	$40K flat 1-year deal. No escalation. Resets each season.
Free agent deal	Bird rights determine max years and escalation rate (8% Bird, 5% non-Bird).

8. Team & Owner System

Each Team object holds identity, owner personality, roster IDs, financials, standings, and draft pick ownership. Full G League Team objects exist with their own identity and affiliate links.

Owner Attributes

tax_willingness	0.0–1.0. How far above the luxury tax the owner will go.
spending_aggressiveness	0.0–1.0. How quickly they spend to cap vs staying conservative.
patience	0.0–1.0. Affects firing threshold (base 8 + patience×8 = range 8–16 pressure points).
ambition	0.0–1.0. Pushes expectation tier up (≥0.8) or down (≤0.39) from engine baseline.
will_tank	Boolean. Owner immune to pressure during explicit tank seasons.

9. Expectations & Coaching Pressure

Each preseason, the engine sets an expectation tier based on roster strength, adjusted by owner ambition. At season end, actual results are scored and compared against expectations.

Tier	Target Score	Description
title_contender	5–8	Win the championship

deep_run	4–6	Conference finals or better
playoff_team	2–4	Top 6 seed in conference
play_in	0–2	Qualify for play-in (seeds 7–10)
development	-1–1	Win 30–40 games, develop youth
tank	-2–0	Lottery positioning

10. Match Simulation Engine

The match engine uses a full possession-by-possession simulation model. Each possession generates a chain of up to 5–6 events (initiation → non-terminal events → terminal), resolving outcomes using all 53 player attributes weighted against opponent attributes, coaching tactics, fatigue, and contextual factors.

Architecture

MatchEngine.gd	Orchestrates a complete game: builds CoachTactics, runs the possession loop, manages fatigue, calls ShotResolver, accumulates box score, emits signals.
PossessionSim.gd	Simulates a single possession as a full event chain. Coaching philosophy, mismatch hunting, flair plays, and all 53 attributes influence every event.
ShotResolver.gd	Resolves shot attempts: make probability, block checks, foul-on-shot, and-ones, free throw simulation, rebound contest.
RotationManager.gd	Tracks minutes, foul trouble, and fatigue. Manages substitutions from the depth chart. Returns starters in Q4 stretch runs.
CoachTactics.gd	Match-time tactics object: philosophy weights, defensive assignments, mismatch scan, zone/press/pack-the-paint modifiers.
QuickSim.gd	Lightweight background game generator. Produces MatchResult-compatible output using Normal distribution score rolls. ~1ms per game.
Coach.gd	Coach entity: 13 coaching attributes (1–5), style choices, philosophies, depth chart, minutes targets, serialization.

Key Design Decisions

Score distribution	Normal distribution (not Poisson) — basketball scores are high and clustered. NBA mean ≈ 113, σ ≈ 11.
Possessions per game	~100 per game (25 per quarter); team pace_target multiplier adjusts volume. Range: 18–32 per quarter.
Full event chain	Up to 5–6 events per possession: ISO, PNR, PNP, Post Entry, Motion Entry, Transition initiations; Drive Kick, Swing/Skip Pass, Hand-Off, Cut, Screen, OREB non-terminals.

Openness model	Every event produces an openness score (0.0–1.0) for the new ballhandler. Openness drives terminal probability and shot make probability.
Coaching philosophy	5 offensive (Pace & Space, ISO-Heavy, Post Up, Ball Movement, Transition First) and 5 defensive (Man, Zone 2-3, Switching, Pack Paint, Press) philosophies with distinct weight profiles.
Execution quality noise	Low Offensive/Defensive Systems coaches see their philosophy weights drifted toward 1.0. A poor coach running Pace & Space bleeds ISO and post-ups.
Mismatch hunting	After every defensive switch, CoachTactics.scan_mismatches() re-evaluates all matchups. Detected mismatches boost pass weights toward the mismatch target. Gated by Tactical Flexibility.
Fatigue model	Stamina degrades attributes throughout the game at a rate modified by Fitness & Conditioning. RotationManager uses fatigue to trigger subs.
Clutch factor	Composure + Important Matches (hidden) activate in final 2 minutes of close games (margin ≤5) and all of OT.
Flair plays	High Flair players occasionally trigger improvised events. Success gated by Flair - Composure delta. Includes no-look passes, euro steps, alley-oops.
Home court advantage	3.5 expected point boost applied as a flat make-probability modifier injected into CoachTactics.
Simulation fidelity	Full engine for player's games (full event chain + box score). QuickSim for all other games (score roll + sparse box score).
Deterministic replay	Single game_seed seeds all three RNGs at offsets. Stored in MatchResult for exact game replay.

Possession Event Chain

Each possession starts at an initiation and chains through non-terminal events until a terminal fires. Chain depth: 3–6 events, weighted toward 4.

Initiation Types

ISO	Ballhandler goes 1-on-1. Weighted by Ball Handling, Decision Making, Flair vs Perimeter Defense, Defensive IQ.
Pick & Roll	Screener sets screen; resolver decides whether ball stays with handler or goes to the roller.
Pick & Pop	Screener pops to the three-point line. Requires good passer and shooter to work effectively.
Post Entry	Ball entered to post player. Entry pass can be deflected/stolen by a high Pass Perception defender.
Motion Entry	Ball moved to wing off off-ball movement. Default philosophy for Ball Movement coaches.
Transition	Fast break off steal or defensive rebound. Flagged by the previous possession's result.

Non-Terminal Events

Drive & Kick	Ballhandler penetrates and kicks to open teammate. Can turnover on a strip.
Swing Pass	Perimeter ball movement. Can be intercepted by high Pass Perception defenders.
Skip Pass	Cross-key pass to weak-side shooter. Higher turnover risk; creates larger openness window on success.
Hand-Off	Dribble hand-off to a cutter. Can be jumped by a high-Anticipation defender.
Cut	Off-ball player cuts to basket. Passer must 'see' the cut (Pass Vision threshold). Missed cuts leave openness unchanged.
Screen	Off-ball screen frees a shooter/cutter. If defense switches: assignments update, mismatch scan fires immediately.
OREB	Second-chance possession. Offensive vs defensive rebounding contest replaces standard initiation.

Shot Resolution

Make probability	base_rate(shot_type) + skill_mod + openness_mod + contest_mod + def_iq_mod + fatigue_mod + clutch_mod + coaching_mod + shot_iq_mod. Clamped 0.05–0.95.
Base rates	Layup 60%, Driving Dunk 72%, Standing Dunk 75%, Close Shot 52%, Post Hook 50%, Post Fade 44%, Mid-Range 42%, Three-Point 36%, Free Throw 78%.
Block check	Runs before make roll on rim attempts. Blocker found from full defensive five (help rotations matter). Wide open shots nearly unblockable.
Foul on shot	Runs independently of make/miss. Draw Foul attr + contest tightness + defender Defensive Consistency. And-one or free throws based on make/miss.
Zone modifier	Zone 2-3 leaves corners open (+openness on corner threes) and packs paint (−make prob on interior shots).
Paint protection	Pack the Paint reduces interior make probability; deliberately concedes perimeter shots.

MatchResult Structure

Both MatchEngine and QuickSim return an identical MatchResult dictionary. The 'simulated' flag distinguishes the source.

home/away_score	Final scores
home/away_quarters	Quarter-by-quarter scores (4 entries + OT)

player_stats	Dictionary: player_id → full stat line (pts, reb, ast, stl, blk, to, fouls, fgm/fga, fg3m/fg3a, ftm/fta, minutes, plus_minus)
home/away_team_stats	Team aggregates: turnovers, oreb, dreb, fouls, fg_pct, off_rating
play_by_play	Array of structured event entries (empty for QuickSim). Each entry has: quarter, clock, team, event type, player ids, result, running score, text.
simulated	false = full MatchEngine result. true = QuickSim result.
overtime_periods	Number of overtime periods played (0 = regulation)
game_seed	RNG seed for exact replay

11. Season Loop & Schedule

Schedule Generation (ScheduleGenerator.gd)

Runs once at new game creation or season rollover. Builds the full 82-game calendar from the NBA scheduling formula, then places games onto dates using a greedy constraint solver.

NBA Scheduling Formula (82 games per team)

Division (16 games)	4 opponents × 4 games each
Intra-conference (36 games)	6 non-division conference opponents × 4 games + 4 opponents × 3 games
Inter-conference (30 games)	15 opposite-conference opponents × 2 games (1 home, 1 away)
Total	1,230 games across all 30 teams

Calendar Placement Constraints

Season window	October 20 → April 10 (~172 days)
Max games per day	15 games on busy nights
Back-to-back limit	Maximum 2 back-to-backs per team per 7-day window
Rolling week limit	Maximum 5 games per team in any 7-day window
Constraint relaxation	Last 5 placement attempts drop the back-to-back check to ensure all 82 games are placed
Marquee dates	Opening Night (Oct 20) and Christmas Day (Dec 25, 5 games) pinned before greedy placement

Season Scheduler (SeasonScheduler.gd)

The daily loop. Called by GameManager when the player advances time. Three entry points:

process_day()	Processes all games on the current WorldState date. Returns a MatchdayResult.
advance_to_next_game()	Drives the clock forward day-by-day until the player's next game. Silently QuickSims background days.
simulate_rest_of_season()	Forces QuickSim on all remaining games including the player's team. Used for fast-forward.

Game Routing

Player's game	Routes through MatchEngine. Full possession-by-possession simulation with play-by-play log.
All other games	Routes through QuickSim. Score roll + sparse box score. ~1ms per game.
Ordering	Background games simulate first so scores from around the league are visible before the player's game.

Weekly Maintenance

Standings rebuild	Every 7 days. WorldState.rebuild_standings() sorts by win_pct with point_diff tiebreaker.
Trade deadline	Feb 6 calendar check fires notification. Trade freeze enforced by TradeEngine (planned).
Injury tick	Deferred to InjuryEngine.tick_week() (planned).
Player dev tick	Deferred to PlayerDev.tick_week() (planned).

Clock.gd Updates

Leap year fix	_days_in_month() and _is_leap_year() added. February now correctly handles 2028, 2032, etc.
is_match_day()	Replaced day-of-week heuristic with live schedule scan. Returns true only if WorldState.schedule has unplayed games today.
Play-in milestone	PLAY_IN_START (Apr 13) added. PLAYOFFS_START corrected to Apr 19 (matches nba_calendar.json).
Stage transitions	All season stage changes now guard against double-firing. Free agency calls WorldState.begin_offseason().

12. League Loader & World Generation

LeagueLoader.gd is a one-shot system called by GameManager.new_game(). It reads all JSON configs, constructs all Team and Player objects, generates staggered contracts, enforces cap compliance, and saves everything to the DB in batch transactions.

World Gen Sequence

1	Load all JSON configs into WorldState.league_configs
2	Set per-league salary caps from config
3	Initialise TV spike schedule (random year within 8–10 season window)
4	Build 30 NBA Team objects from nba_teams.json
5	Build 30 G League Team objects from g_league_teams.json
6	Assign 5 years of draft picks (both rounds) to all 30 NBA teams
7	Generate NBA rosters: 15 players × 30 teams = 450 players
8	Generate G League rosters: 10 players × 30 teams = 300 players
9	Generate draft class: 60 prospects for next draft
10	Generate staggered contracts for all ~810 players
11	Two-pass cap compliance: trim over-apron, bump under-floor
12	Batch save all players, teams, and contracts to DB
13	Start player_tenure records for all rostered players
14	Rebuild initial standings (all 0–0)
15	Generate 82-game season schedule via ScheduleGenerator and save to DB

13. Development Roadmap

Feature	Description	Status
Godot 4 project setup	Project structure, folders, settings	**Done**
SQLite plugin	godot-sqlite via AssetLib, schema created	**Done**
Autoload singletons	EventBus, Database, WorldState, Clock, GameManager	**Done**
Player entity	53 attributes across 7 categories, scout visibility, delta potential	**Done**
PlayerGenerator	Tier system, archetype system, position biases, age degradation	**Done**
Team entity	Cap helpers, roster management, pick protections, standings, pressure system	**Done**
Database layer	Full schema: 53 attrs, player_tenure, contracts, teams, picks, match_results	**Done**
League JSON configs	5 NBA files + g_league_teams.json — cap, calendar, teams, expectations	**Done**
ContractGenerator	World gen, free agency, draft night, G League — staggered + bird rights	**Done**

LeagueLoader	One-shot world gen: 810 players, 60 teams, contracts, tenure, standings	**Done**
WorldState	Cap system, TV spike tracking, standings rebuild, typed accessors, roster helpers	**Done**
Coach entity	13 coaching attributes, style/personality creation, philosophy enums, serialization	**Done**
CoachTactics	Match-time tactics: philosophy weights, defensive assignments, mismatch scanning, zone/press helpers	**Done**
PossessionSim	Full event chain (5–6 events): 6 initiations, 7 non-terminals, flair plays, openness model	**Done**
ShotResolver	Shot math, block check, foul-on-shot, and-ones, free throws, rebound resolution	**Done**
RotationManager	Minutes tracking, foul trouble, fatigue subs, Q4 starters return, depth chart execution	**Done**
MatchEngine	Game orchestrator: CoachTactics build, possession loop, fatigue, box score accumulation	**Done**
QuickSim	Lightweight background game generator: Normal distribution scores, sparse box score	**Done**
ScheduleGenerator	82-game season calendar from NBA formula. Greedy constraint placement with back-to-back limits	**Done**
SeasonScheduler	Daily season loop: game routing, advance_to_next_game, weekly tasks, season completion	**Done**
EventBus updates	matchday_complete, regular_season_complete signals added	**Done**
Clock.gd updates	Leap year fix, schedule-aware is_match_day(), play-in milestone, stage transition guards	**Done**
CommentaryEngine	Event → human-readable play-by-play text. Reads structured play_events log.	**Planned**
PlayoffEngine	Play-in tournament, bracket generation, series simulation	**Planned**
Draft engine	Prospect generation, scouting, lottery, draft night UI	**Planned**
Trade engine	AI evaluation, salary matching, pick trades, trade UI	**Planned**
Free agency system	Player decisions, bidding, two-way contracts	**Planned**
Player development	Aging curves, training, potential reveal	**Planned**
Injury engine	Injury probability, severity, recovery timeline	**Planned**
AI Coach generator	Generates coaches with matching philosophies for all 30 AI teams	**Planned**

Inbox / news engine	FM-style message inbox: contract offers, scout reports, board expectations	**Planned**
Hub UI screen	Main dashboard, inbox, calendar, quick stats, scores ticker	**Planned**
Squad / tactics screen	Roster management, lineup setting, rotations, depth chart editor	**Planned**
Match day screen	Text play-by-play, live score, substitutions, coaching decisions, team talks	**Planned**
Career mode logic	Job offers, reputation system, coach progression, owner expectations	**Planned**
Scouting system	Scout ratings, report accuracy, overseas scouting	**Planned**
Save / load system	Full game state serialization via SQLite	**Planned**
G League season loop	50-game season, own standings, promotion/call-up pipeline	**Planned**
College pipeline	Recruiting, 4-year development, draft entry	**Planned**
Overseas leagues	EuroLeague config, international transfers	**Planned**

14. Key References

Football Manager	Attribute simulation, reputation system, text match engine, scouting inaccuracy, inbox system, simulation fidelity tiers
OOTP Baseball	Current + potential dual ratings, financial coefficient, league config flexibility
NBA 2K (attribute model)	53-attribute model adapted from 2K's attribute categories and archetype system
OpenFootManager (GitHub)	Open source manager: Rust sim engine + React UI. Architecture separation pattern.
open-source-ball (GitHub)	Swift iOS college hoops sim. College pipeline logic, chunked save approach.
xg_simulator.py (Arya)	Existing Poisson/Monte Carlo match sim. Informs possession engine design philosophy.

15. Design Notes & Decisions

Licensing

All teams and players ship with real NBA names for development. Architecture supports community database mods to swap names if needed. Fictional names are a deferred decision.

No 3D Engine (by design)

The 3D match view is explicitly out of scope for v1. Text-based simulation with rich play-by-play commentary is the target. A 2D match view may be added later as a non-blocking feature.

Coaching Philosophy Execution Quality

A poor Offensive Systems coach (rating 1) running Pace & Space sees their weight multipliers pulled 75% toward 1.0 with random variance on top. Their team tries to run it but ISO and post-ups bleed through. An elite coach's weights are clean and accurate to intent. This is the difference between Popovich's system and a first-year G League coach installing the same offense.

GDScript vs C#

GDScript chosen for Python-like syntax matching existing background and tight Godot integration. If performance becomes an issue, the simulation engine can be ported to C# without touching UI code due to clean architecture separation.

Deterministic Replay

Every game is seeded from a single integer stored in MatchResult.game_seed. All three RNGs (MatchEngine, PossessionSim, ShotResolver) are seeded at offsets from this value. Replaying a game from the same seed produces identical results.

Known Issue — World Gen Roster IDs

Roster IDs on Team objects are populated during player generation before DB save assigns real auto-incremented IDs. Requires a post-save ID fixup pass. To be resolved during first Godot integration test.